

Homework #1 Solution

For the questions below, use the following constants:

Speed of light: $c = 2.998 \times 10^8$ m/s

Electron charge: $e = 1.602 \times 10^{-19}$ C

Plank constant: $h = 6.626 \times 10^{-34}$ J/Hz

- 1- (a) Wavelength range is from 0.4 to 0.7 μm .

$$f_1 = \frac{c}{\lambda_1} = \frac{3 \times 10^8 \text{ m/s}}{0.4 \mu\text{m}} \approx 750 \text{ THz}$$

$$f_2 = \frac{c}{\lambda_2} = \frac{3 \times 10^8 \text{ m/s}}{0.7 \mu\text{m}} \approx 429 \text{ THz}$$

Therefore, the corresponding frequency range is 429 ~ 750 THz.

- (b) 1 Mbps: $T = \frac{1}{R_b} = 1 \mu\text{s}$, $E = TP = 1 \mu\text{s} \times 1 \text{ mW} = 1 \text{ nJ}$

$$1 \text{ Gbps: } T = \frac{1}{R_b} = 1 \text{ ns}, E = TP = 1 \text{ ns} \times 1 \text{ mW} = 1 \text{ pJ}$$

- (c) Loss is $10^5 \text{ dB/km} \times 5 \text{ mm} = 0.5 \text{ dB}$

$$\text{Fraction is } 10^{-0.05} \approx 0.89 = 89\%$$

- (d) $v = \frac{c}{n} \approx 1.998 \times 10^8$ m/s

- 2- Fiber loss: $50 \text{ km} \times 0.5 \text{ dB/km} = 25 \text{ dB}$

$$\text{Fiber connectors: } \left\lfloor \frac{50}{3} \right\rfloor \times 0.3 = 4.8 \text{ dB}$$

Tx to fiber: 1 dB

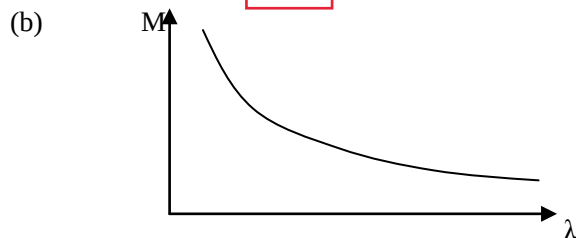
Rx to fiber: 1 dB

Total loss is 31.8 dB

Tx power is 1 mW = 0 dBm

Received power is 0 dBm – 31.8 dB = -31.8 dBm

- 3- (a) $V \approx 83.23$, $M \approx 53$ similar numbers are OK)



- 4- $\theta = \arcsin\left(\sin 45^\circ \times \frac{1.5}{1.33}\right) \approx 52.9^\circ$

- 5- $\theta_c = \arcsin\left(\sin 90^\circ \times \frac{1.33}{1.5}\right) \approx 62.46^\circ$, only when light goes from high index media (glass in this question) to low index media (water in this question).

6- (a) $f = \frac{c}{\lambda} \approx 193.6 \text{ THz}$

(b) $df = -\frac{c}{\lambda^2} d\lambda$. When $d\lambda$ is 0.1 nm, df is about 12.5 GHz.

7- (a) Energy of a photon is $E = hf = hc/\lambda$, $1 \text{ J} = 6.24 \times 10^{18} \text{ eV}$

0.82 μm : 1.51 eV

1.3 μm : 0.955 eV

1.55 μm : 0.801 eV

(b) $\lambda = hc/E$

Emitted wavelength in vacuum: $\lambda_{\text{emit}} = 1.0 \mu\text{m}$

Frequency in fiber is the same with frequency in vacuum: $f = c/\lambda_{\text{emit}} = 299.8 \text{ THz}$

8- (a) The wavelength of mode should satisfy $m\lambda = 2nL$, where m is an integer.

$$m_1 = \left\lfloor \frac{2nL}{\lambda_1} \right\rfloor = 2023$$

$$m_2 = \left\lfloor \frac{2nL}{\lambda_2} \right\rfloor = 1972$$

therefore the number of modes is $m_1 - m_2 + 1 = 2023 - 1972 + 1 = \underline{52}$.

(b) calculating in a similar way, $m_1 = 2$, $m_2 = 2$, so the number of modes is $m_1 - m_2 + 1 = \underline{1}$

9- (a) The time Ray B passing through a fiber with a length of L is

$$t_B = \frac{L}{v} = \frac{Ln_1}{c}$$

The distance that Ray A propagates in a fiber with a length of L is

$$L_A = \frac{L}{\sin\phi} = \frac{Ln_1}{n_2}$$

The time Ray A passing through a fiber with a length of L is

$$t_A = \frac{L_A}{v} = \frac{Ln_1^2}{cn_2}$$

Pulse spreading is

$$\Delta T = t_A - t_B = \frac{Ln_1(n_1 - n_2)}{cn_2}$$

(b) Use the equation in (a)

$$\Delta T = \frac{1000 \times 1.475(1.475 - 1.460)}{2.998 \times 10^8 \times 1.460} \approx 50.55 \text{ ns}$$

(c) Number of modes is

$$M = \left\lfloor \frac{2 \times 25 \times 10^{-6} \sqrt{1.475^2 - 1.460^2}}{1.55 \times 10^{-6}} \right\rfloor = 6$$